EE115 Analog Circuits
模拟电路

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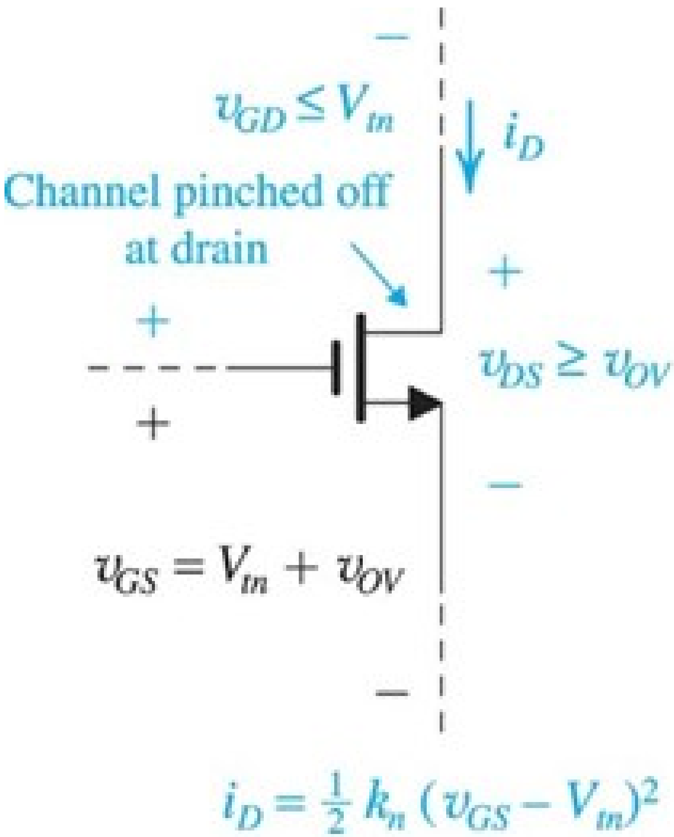
Outline

- Transistor amplifier 1
- SEDTRA/SMITH book page 367-393；

Transistor Operating Mode in Amplifiers



(a)

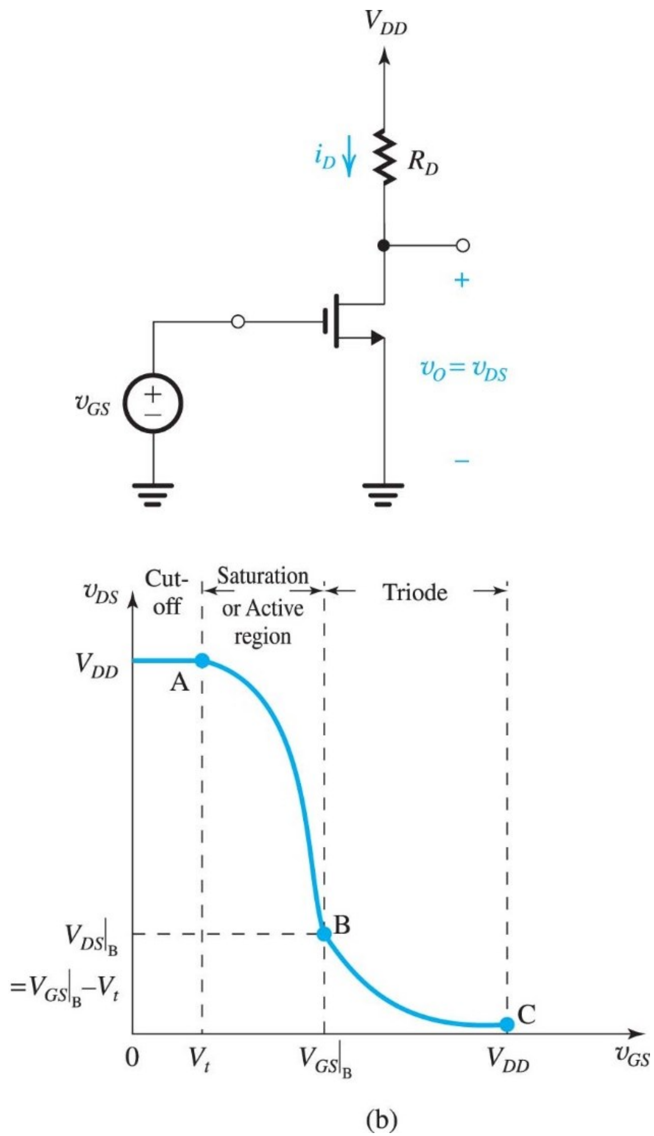
Transistors are biased in flat part of the I-V curves

- Saturation mode for MOSFET

  Channel pinched off at drain

- Drain voltage can vary without changing current

Voltage Transfer Curves



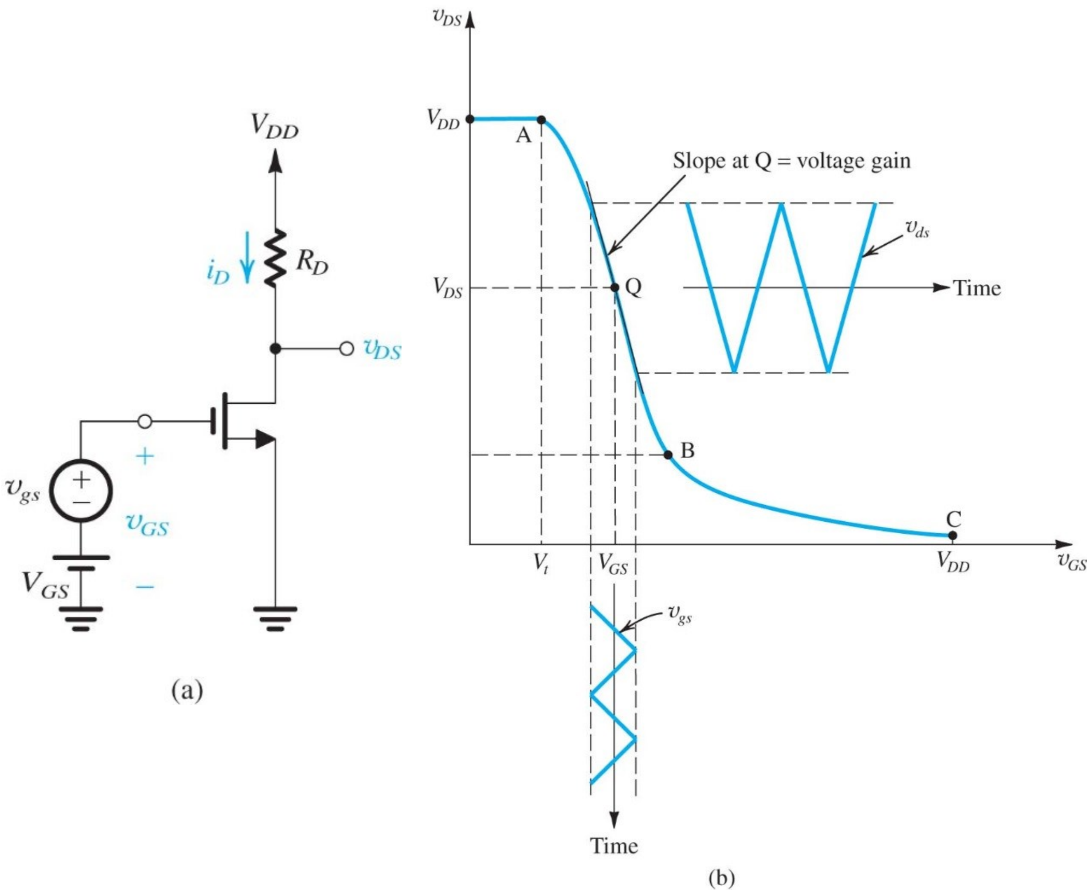
- Transistors are transconductance amplifiers
  - MOSFET: v_{GS} controls i_D

- Adding load resistors converts current to voltage -> voltage amplifiers
  - Drain voltage free to change in saturation mode

$$v_{DS} = V_{DD} - i_D R_D$$

Figure 7.2 (**a**) An NMOS amplifier and (**b**) its VTC; and (**c**) an *npn* amplifier and (**d**) its VTC.

Biasing the amplifier



$$v_{DS} = V_{DD} - \frac{1}{2} k_n R_D (v_{GS} - V_t)^2$$

$$v_{GS}(t) = V_{GS} + v_{gs}(t)$$

$$A_v = \left. \frac{dv_{DS}}{dv_{GS}} \right|_{v_{GS} = V_{GS}}$$

Figure 7.4 The MOSFET amplifier with a small time-varying signal $v_{gs}(t)$ superimposed on the dc bias voltage V_{GS}. The MOSFET operates on a short almost-linear segment of the VTC around the bias point Q and provides an output voltage $v_{ds} = A_v v_{gs}$

Example

Example 7.1

Consider the amplifier circuit shown in Fig. 7.4(a). The transistor is specified to have $V_t = 0.4$ V, $k_n' = 0.4$ mA/V^2, $W/L = 10$, and $\lambda = 0$. Also, let $V_{DD} = 1.8$ V, $R_D = 17.5$ kΩ, and $V_{GS} = 0.6$ V.

(a) For $v_{gs} = 0$ (and hence $v_{ds} = 0$), find V_{OV}, I_D, V_{DS}, and A_v.
(b) What is the maximum symmetrical signal swing allowed at the drain? Hence, find the maximum allowable amplitude of a sinusoidal v_{gs}.

Solution

(a) With $V_{GS} = 0.6$ V, $V_{OV} = 0.6 - 0.4 = 0.2$ V. Thus,

$$\begin{aligned} I_D &= \frac{1}{2} k_n' \left(\frac{W}{L}\right) V_{OV}^2 \\ &= \frac{1}{2} \times 0.4 \times 10 \times 0.2^2 = 0.08 \text{ mA} \\ V_{DS} &= V_{DD} - R_D I_D \\ &= 1.8 - 17.5 \times 0.08 = 0.4 \text{ V} \end{aligned}$$

Since V_{DS} is greater than V_{OV}, the transistor is indeed operating in saturation. The voltage gain can be found from Eq. (7.15),

$$\begin{aligned} A_v &= -k_n V_{OV} R_D \\ &= -0.4 \times 10 \times 0.2 \times 17.5 \\ &= -14 \text{ V/V} \end{aligned}$$

An identical result can be found using Eq. (7.17).

(b) Since $V_{OV} = 0.2$ V and $V_{DS} = 0.4$ V, we see that the maximum allowable negative signal swing at the drain is 0.2 V. In the positive direction, a swing of $+0.2$ V would not cause the transistor to

Continued

Example 7.1 *continued*

cut off (since the resulting v_{DS} would be still lower than V_{DD}) and thus is allowed. Thus the maximum symmetrical signal swing allowable at the drain is ± 0.2 V. The corresponding amplitude of v_{gs} can be found from

$$\hat{v}_{gs} = \frac{\hat{v}_{ds}}{|A_v|} = \frac{0.2 \text{ V}}{14} = 14.2 \text{ mV}$$

Since $\hat{v}_{gs} \ll V_{OV}$, the operation will be reasonably linear (more on this in later sections).

Greater insight into the issue of allowable signal swing can be obtained by examining the signal waveforms shown in Fig. 7.5. Note that for the MOSFET to remain in saturation at the negative peak of v_{ds}, we must ensure that

$$v_{DS\min} \geq v_{GS\max} - V_t$$

that is,

$$0.4 - |A_v|\hat{v}_{gs} \geq 0.6 + \hat{v}_{gs} - 0.4$$

which results in

$$\hat{v}_{gs} \leq \frac{0.2}{|A_v| + 1} = 13.3 \text{ mV}$$

This result differs slightly from the one obtained earlier.

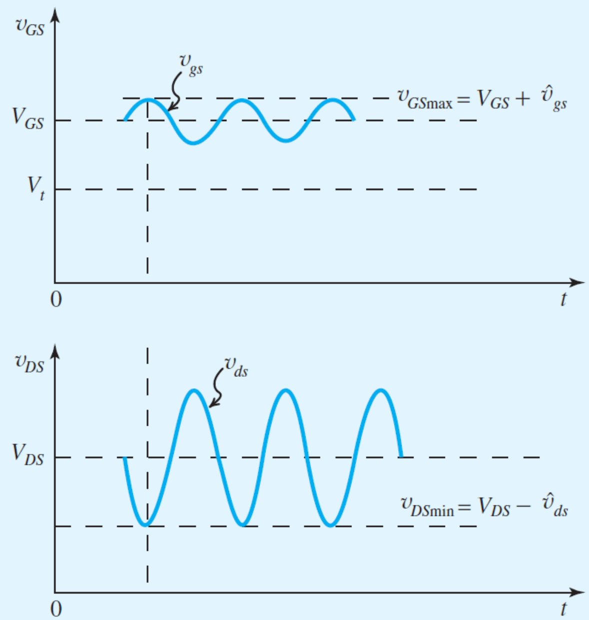
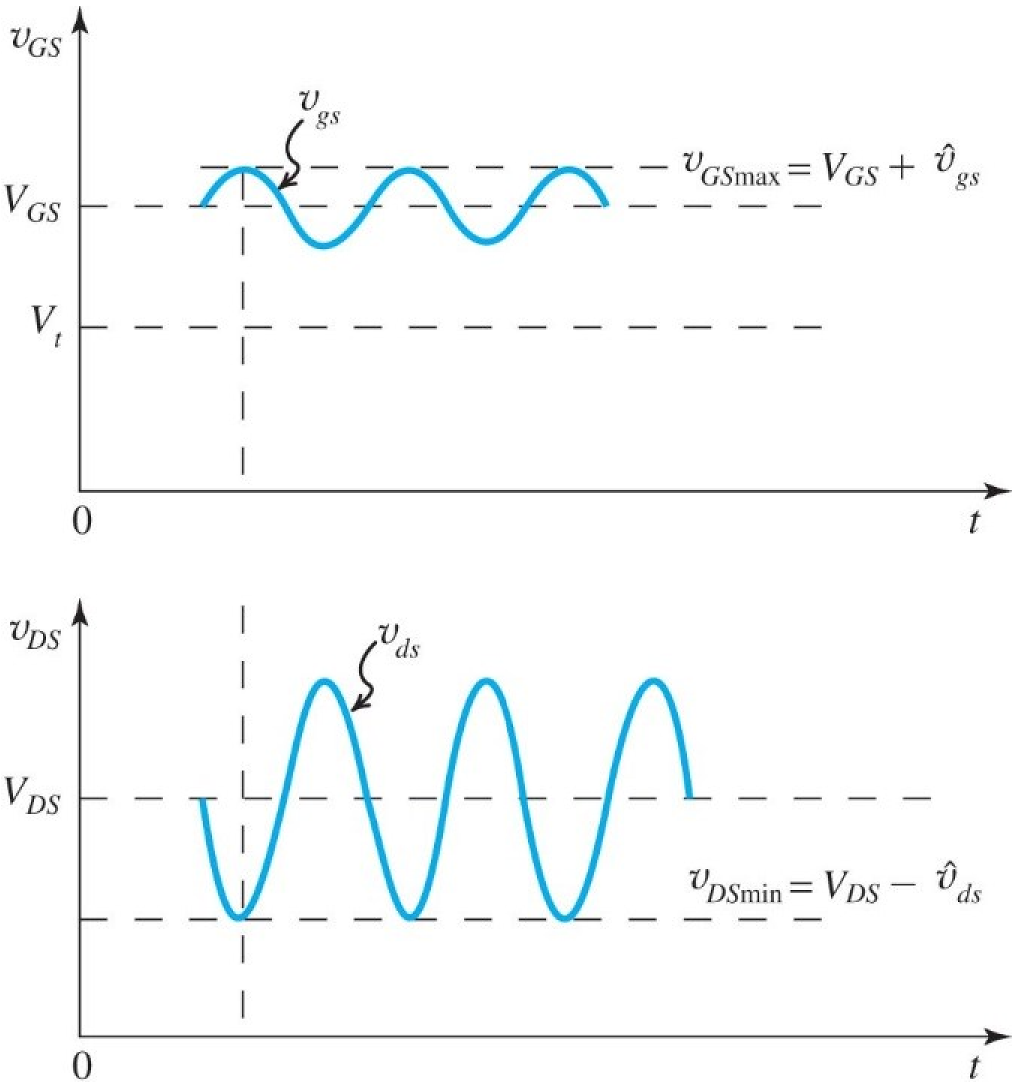


Figure 7.5 Signal waveforms at gate and drain for the amplifier in Example 7.1. Note that to ensure operation in the saturation region at all times, $v_{DS\min} \geq v_{GS\max} - V_t$.

Maximum Input Signal Amplitude

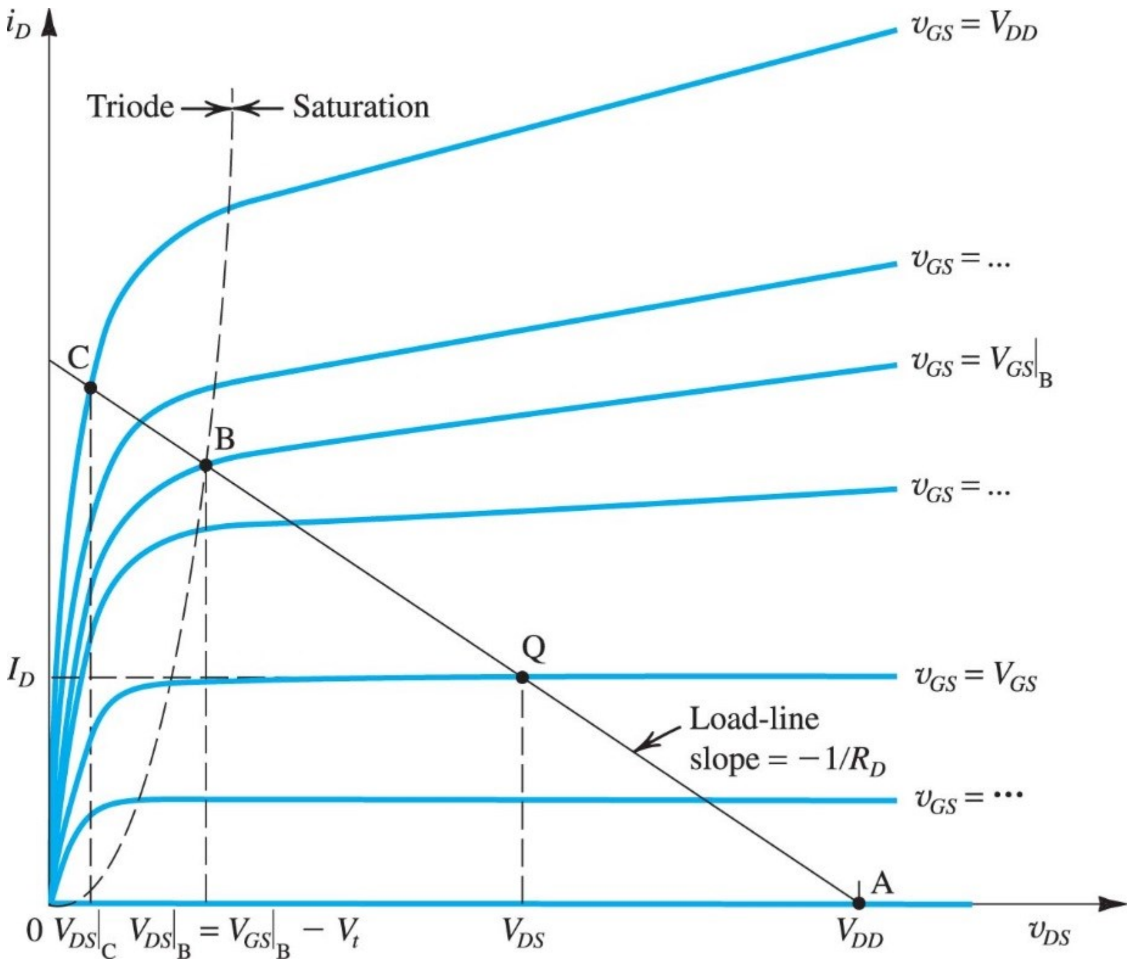


To keep the MOSFET in saturation region:

$$v_{DS\min} \geq v_{GS\max} - V_t$$

Figure 7.5 Signal waveforms at gate and drain for the amplifier in Example 6.1. Note that to ensure operation in the saturation at all times, $v_{DSmin} \geq v_{GSmax} - V_t$

Graphical Analysis with Load Line

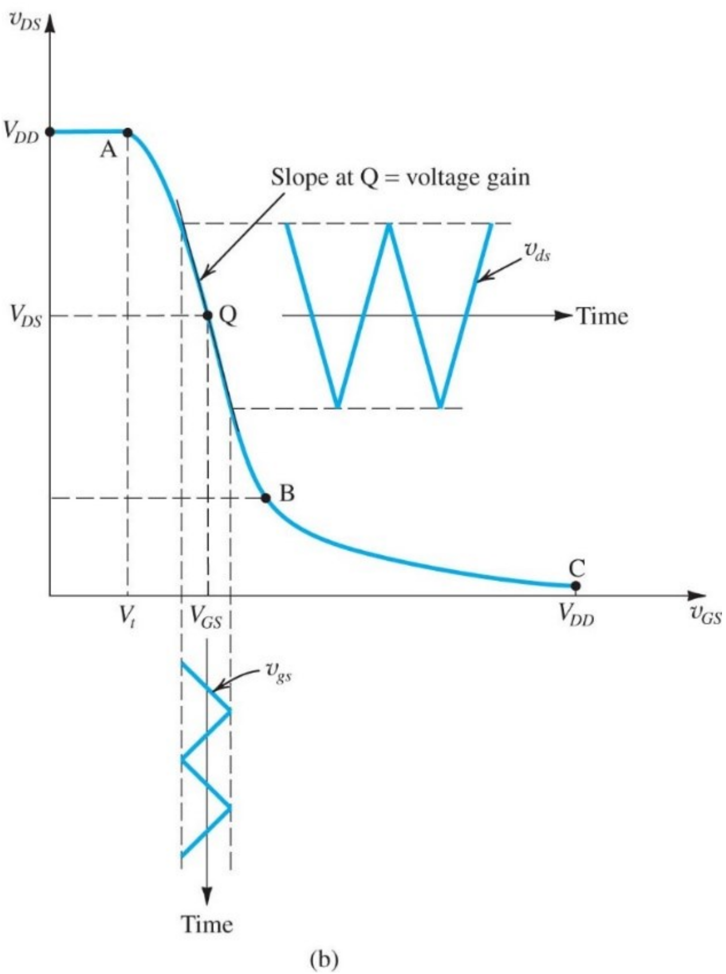


Load line:

$$i_D = \frac{V_{DD}}{R_D} - \frac{1}{R_D} v_{DS}$$

Figure 7.7 Graphical construction to determine the voltage-transfer characteristic of the amplifier in Fig. 7.4(a).

Consideration for Bias Point



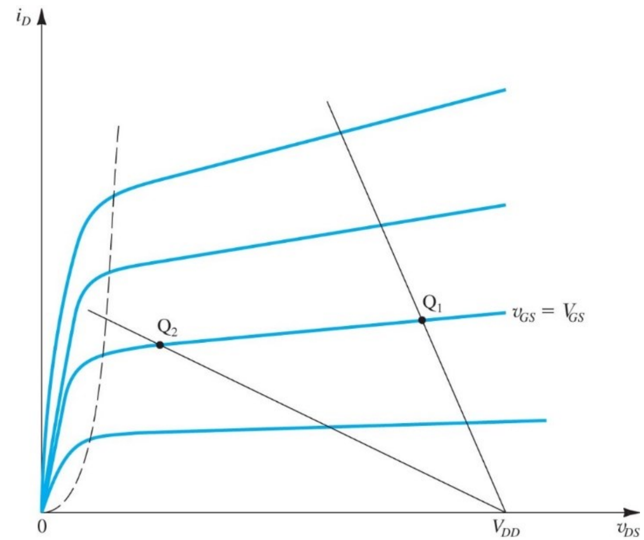


Figure 7.9 Two load lines and corresponding bias points. Bias point Q_1 does not leave sufficient room for positive signal swing at the drain (too close to V_{DD}). Bias point Q_2 is too close to the boundary of the triode region and might not allow for sufficient negative signal swing.

$$i_D = \frac{V_{DD}}{R_D} - \frac{1}{R_D} v_{DS}$$

Small signal model for MOSFET

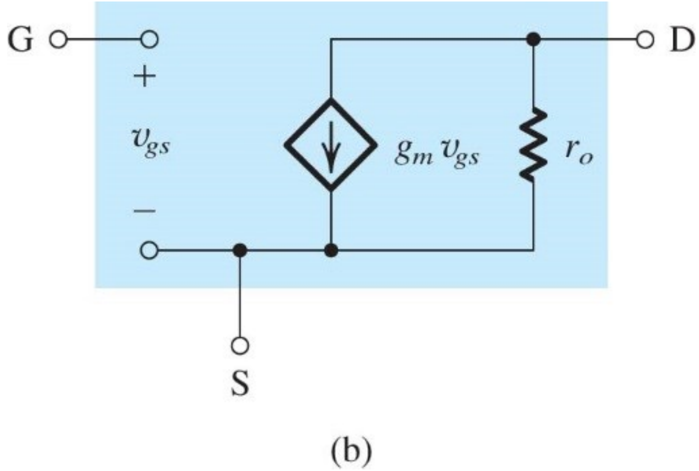


Figure 7.13 Small-signal models for the MOSFET: (b) including the effect of channel-length modulation, modeled by output resistance $r_o = |V_A|/I_D$. These models apply equally well for both NMOS and PMOS transistors.

- The equivalent circuit is valid for both NMOS and PMOS
- In PMOS, use absolute sign for all parameters: $|V_{GS}|$, $|V_t|$, $|V_{OV}|$, $|V_A|$, and replace k_n with k_p

Transconductance:

$$g_m \equiv \left.\frac{\partial i_D}{\partial v_{GS}}\right|_{v_{GS}=V_{GS}}$$

$$g_m \equiv \frac{i_d}{v_{gs}} = k_n(V_{GS} - V_t)$$

$$g_m = k_n V_{OV}$$

Vov= $\sqrt{2I_D/(k_n'(W/L))}$

$$g_m = \sqrt{2k_n'}\sqrt{W/L}\sqrt{I_D}$$

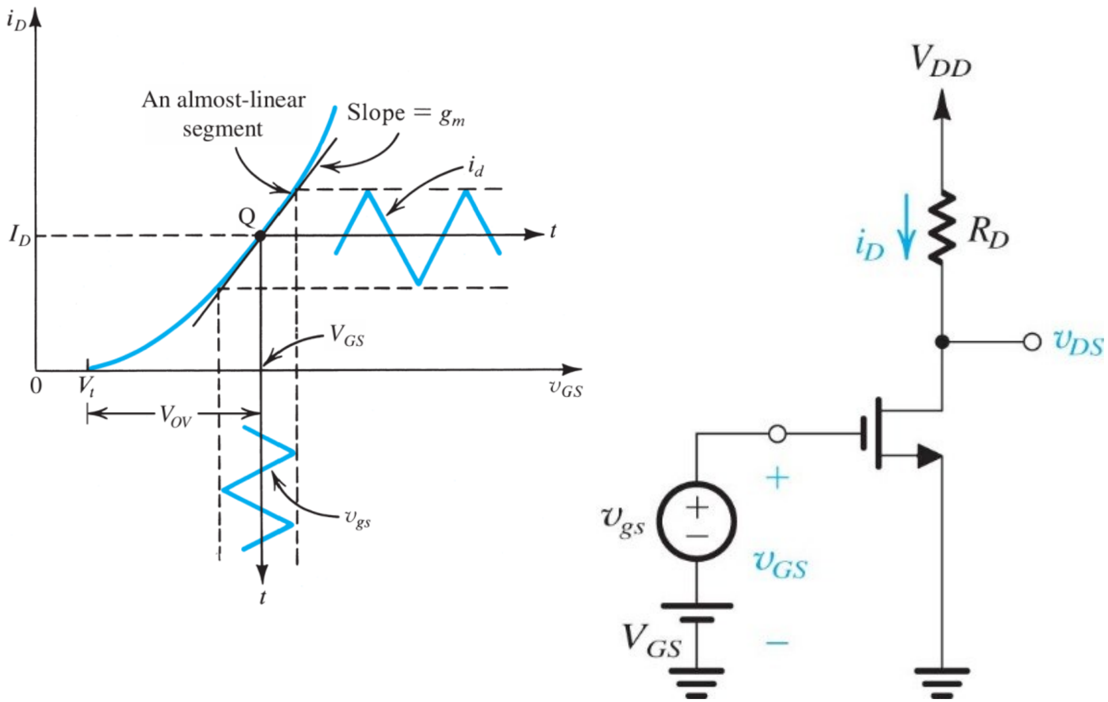
Output resistance at drain:

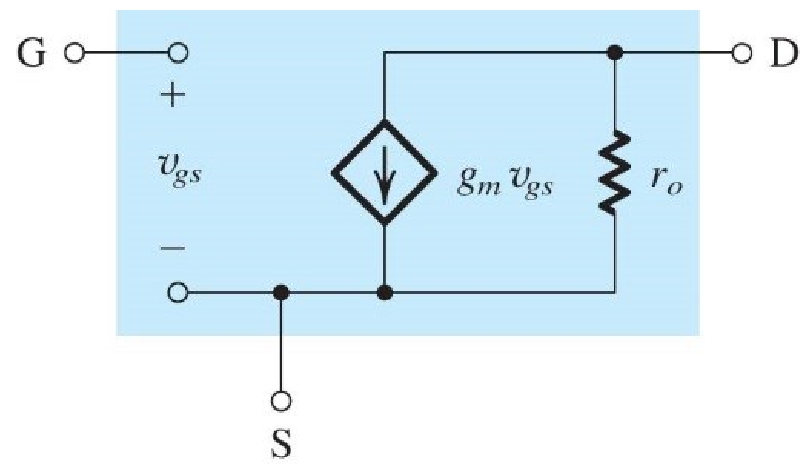
$$r_o = \frac{|V_A|}{I_D}$$

$$i_D = \frac{1}{2}k_n\left(V_{GS} + v_{gs} - V_t\right)^2$$
$$= \frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$$

$$i_d = k_n(V_{GS} - V_t)v_{gs}$$

Applying small-signal model





(b)

Figure 7.10 Conceptual circuit utilized to study the operation of the MOSFET as a small-signal amplifier.

$$i_D = \frac{1}{2}k_n\left(V_{GS} + v_{gs} - V_t\right)^2$$

$$= \frac{1}{2}k_n(V_{GS} - V_t)^2 + k_n(V_{GS} - V_t)v_{gs} + \frac{1}{2}k_n v_{gs}^2$$

Systematic Procedure for Transistor Amplifier Analysis

- 1. Perform DC analysis (ignore small signal source)
- 2. Calculate small-signal parameters (g_m, r_o, etc)
- 3. Generate AC small-signal equivalent circuit
  - » Replace DC voltage source by short circuit
  - » Replace DC current source by open circuit
  - » Replace transistor by hybrid-π model (or other model)
- 4. Perform circuit analysis to determine voltage gain or other amplifier performance parameters

MOSFET Amplifier Example

Given: The MOSFET has V_t = 1.5V, k_n = 0.25 mA/V and Early Voltage = 50V. Find voltage gain for the amplifier.

(1) Solve DC Bias Point

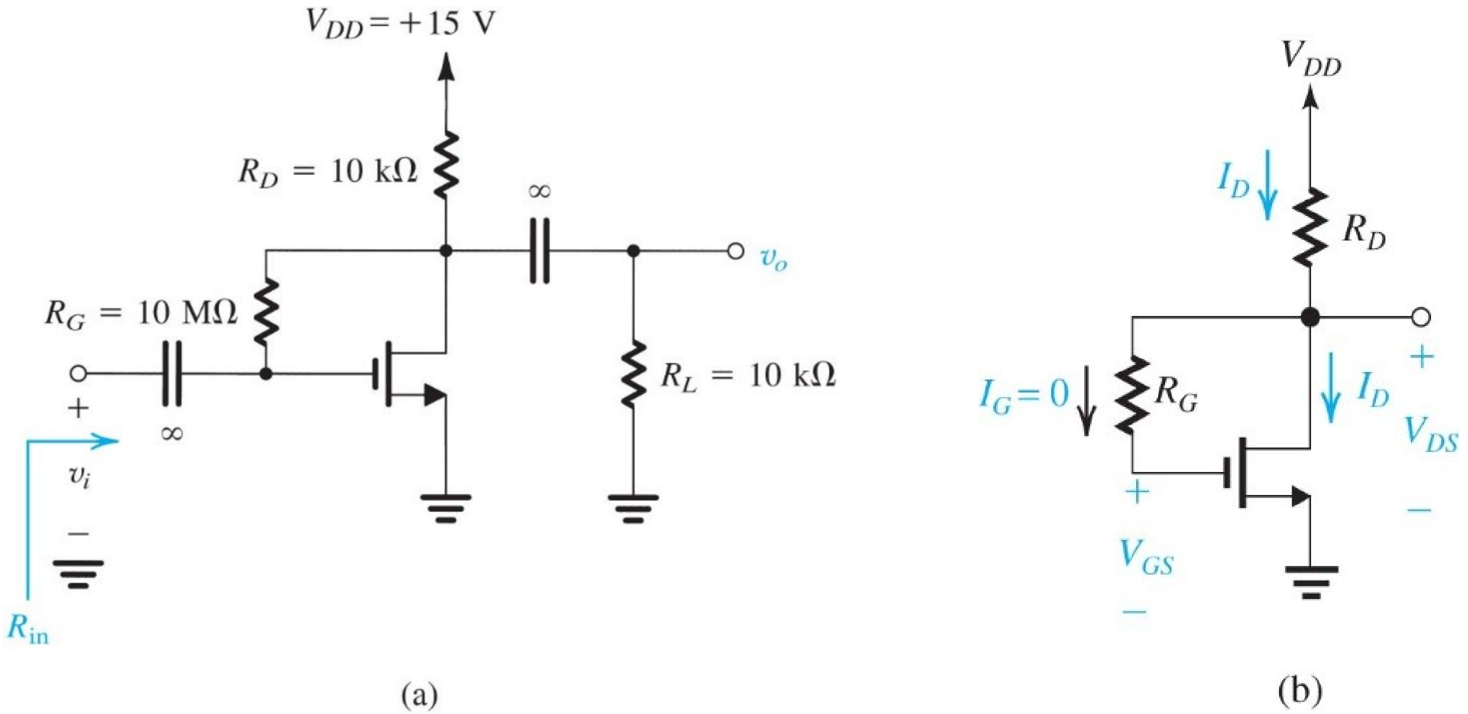


Figure 7.15 Example 7.3: **(a)** amplifier circuit; **(b)** circuit for determining the dc operating point;

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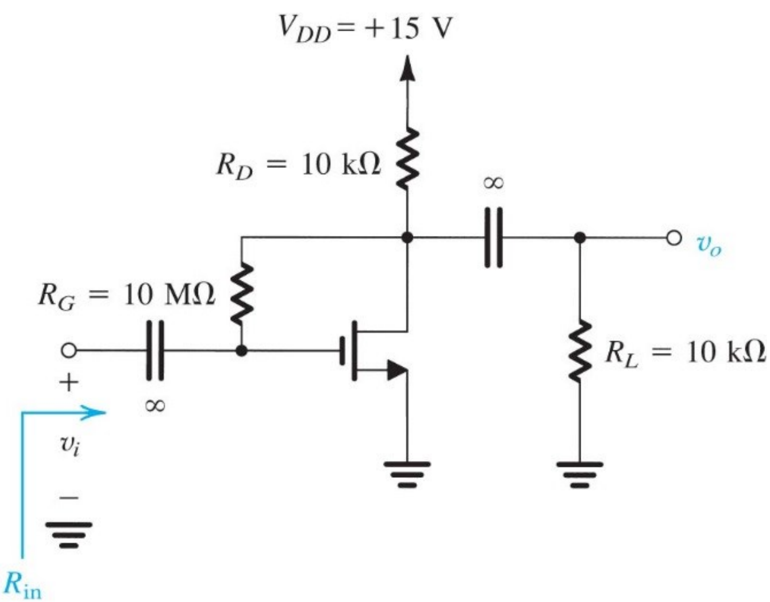
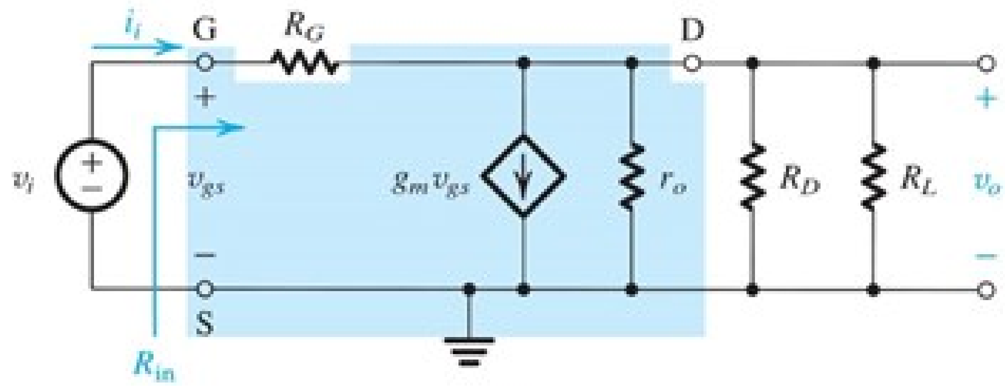
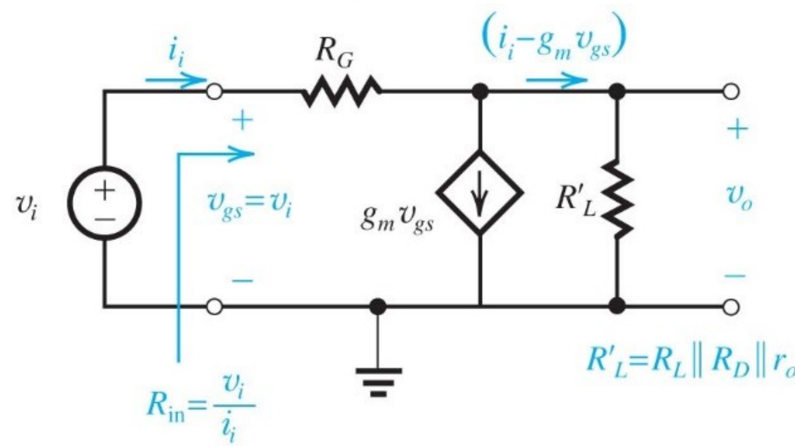
(2) Solve AC Small Signal Circuit



(a)



(c)



(d)

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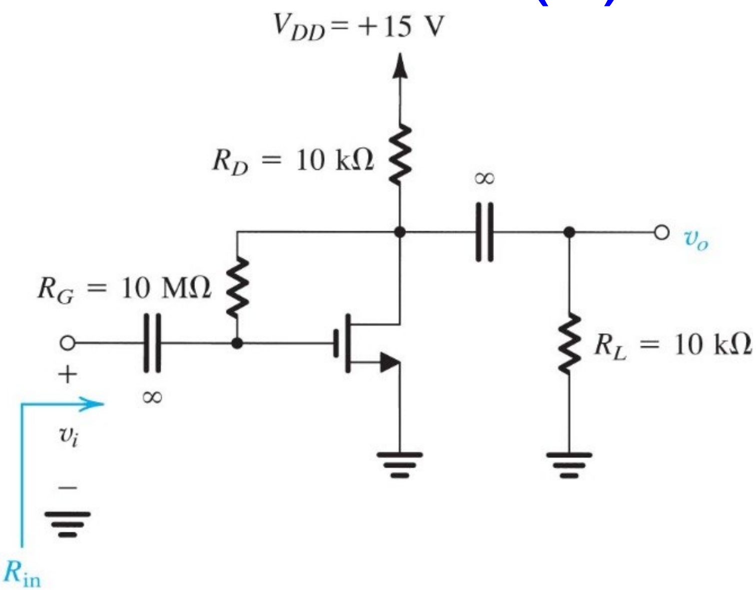
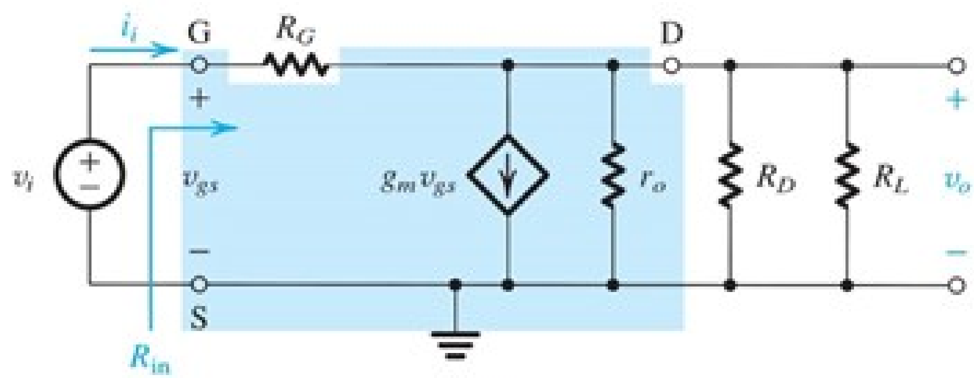
(3) Additional Parameters of Interest



(a)



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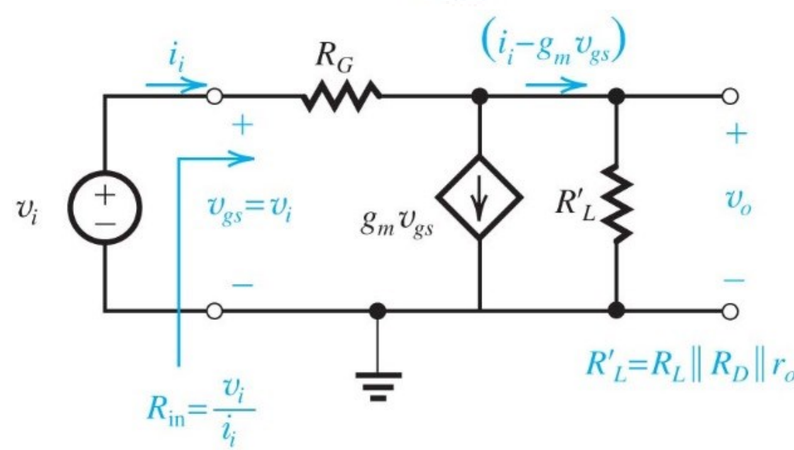

(d)

Homeworkx-due March xx 11p.m.

SEDRA/SMITH Book chapter xx problem xx,

chapter x problem xx